

Tutorial 4

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Alternating Finite Automata

Q1. Prove or disprove: For every DFA D with n states, there exists an AFA A with $O(\log n)$ states recognising the same language.

Q2. Characterise exactly the class of languages recognised by universal automata i.e., every transition is a $\forall$ transition, and prove your characterisation.

Q3. Consider

$$L_n = \{w \in \{0, 1\}^* : |w| \equiv 0 \pmod{p_i} \text{ for all } i = 1, \dots, k\},$$

where $p_1, \dots, p_k$ are the first k primes and

$$\sum_{i=1}^k p_i = n.$$

Show that L_n can be recognised by an AFA with $O(n)$ states, but every NFA for L_n requires $2^{O(n)}$ states.

Q4. MSO and automaton size. Recall that monadic second-order logic (MSO) over finite words extends first-order logic with quantification over sets of positions. A sentence φ defines a language $L(\varphi) \subseteq \Sigma^*$.

- Show that every MSO sentence can be converted into an equivalent DFA. Give an upper bound on the number of states in terms of $|\varphi|$, and show that this bound is a tower of exponentials whose height is $O(|\varphi|)$ and is non-elementary.
- Explain how using alternating finite automata instead of NFAs can reduce the automaton size by one exponential level. Why does this reduce only one level rather than eliminating the tower entirely?
- Conclude that the state complexity of MSO-to-AFA conversion is still non-elementary in $|\varphi|$.

PDAs and CFGs

Q1. For the following languages give context-free grammars to accept them. You must write a 2-3 line justification for each. Also, for each language, give an informal and formal description of a PDA accepting the language. Since for PDA acceptance by empty stack and final state are equivalent, you can use either definition.

- $L_1 = \{w \in \{0, 1\}^* \mid w \text{ contains at least four 1's}\}$
- $L_2 = \{w \in \{0, 1\}^* \mid w \text{ has odd length and middle letter is 0}\}$
- $L_3 = \{w \in \{0, 1\}^* \mid \text{number of 0's in } w \text{ is equal to number of 1's in } w\}$
- $L_4 = \{0^n 1^m 0^n \mid m, n \geq 1\}$

Q2. Consider $L_5 = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i = j \text{ or } k = 2j\}$.

- (a) Construct a PDA to accept L_5 .
- (b) Give a context-free grammar (CFG) that accepts L_5 .
- (c) Prove or disprove: L_5 cannot be accepted by any DPDA.

Q3. (From many states to one state) Consider $L_6 = \{0^n 10^n \mid n \geq 1\}$.

- (a) Give a PDA with three states accepting L_6 .
- (b) Can you give a PDA with just one state (but maybe a larger stack alphabet) which accepts L_6 ?
- (c) Generalize this argument to show why for ANY PDA P accepting L , we can construct another PDA P' which has only one state which accepts L .

Q4. (Acceptance by empty stack in PDA) Prove formally that PDA with acceptance by empty stack are expressively equivalent to PDA with acceptance by final state. Recall that expressively equivalent means that they accept the same set of languages. (The idea of the proof was given in class, this exercise is to ask you to write the details formally!)

Q5. Conversions:

- (a) Convert the grammar $S \rightarrow S0S1S0S \mid S0S0S1S \mid S1S0S0S \mid \varepsilon$ into a PDA that accepts the same language by empty stack.
- (b) Consider $\{a^i b^j c^k \mid i = 2j \text{ or } j = 2k\}$. Construct a CFG for it and then convert it into a PDA.
- (c) Convert the following PDA which accepts by empty stack into a CFG using the algorithm given in lectures. Also simplify the resultant CFG.

$$(\{q\}, \{0, 1\}, \{\perp, A, B\}, \delta, q, \perp, \emptyset)$$

where δ is given by

- $\delta(q, 0, \perp) = (q, A\perp), \quad \delta(q, 1, \perp) = (q, B\perp)$
- $\delta(q, 0, A) = (q, AA), \quad \delta(q, 1, B) = (q, BB)$
- $\delta(q, 1, A) = (q, \varepsilon), \quad \delta(q, 0, B) = (q, \varepsilon)$
- $\delta(q, \varepsilon, \perp) = (q, \varepsilon)$

Q6. (Closure properties) Let L and L' be two context-free languages and let R be a regular language. Recall that a language is context-free if it can be accepted by a PDA or if some CFG accepts it. Show that:

- (a) (concat) $L \circ L'$ is always a context-free language.
- (b) (intersection with regular lang) $L \cap R$ is always a context-free language.